

GitHub & VS Code Simple Guide

Option 1: Initialize Local Folder and Push to GitHub

Step 1: Check Git Configuration

```
git config --global user.name  
git config --global user.email
```

Step 2: Set Git Configuration (if needed)

```
git config --global user.name "Your Name"  
git config --global user.email "your.email@example.com"
```

Step 3: Create .gitignore File (if needed)

Create a `.gitignore` file in your project root:

```
# Python Virtual Environments  
.venv/  
venv/  
env/  
  
# Environment files  
.env  
  
# Python  
__pycache__/  
*.pyc
```

Step 4: Initialize Git in Terminal

1. Open your project folder in VS Code
2. Open terminal (`Ctrl+``` or `View > Terminal`)
3. Run: `git init`

4. Run: `git add .`

5. Run: `git commit -m "Initial commit"`

Step 4: Create GitHub Repository via Terminal (GitHub CLI)

Check if GitHub CLI is installed:

```
gh --version
```

If "command not found" but GitHub CLI is installed:

- **Windows:** Restart VS Code/terminal or check if it's in PATH
- **Mac/Linux:** Try `which gh` or restart terminal
- **Alternative:** Use full path like `/usr/local/bin/gh --version`

If GitHub CLI exists:

```
gh auth login  
gh repo create [REPO_NAME] --public --source=. --remote=origin --push
```

If GitHub CLI not found, install from cli.github.com or use the website method below.

Alternative Step 4: Create Repository on GitHub Website

1. Go to github.com and sign in
2. Click the "+" icon → "New repository"
3. Name your repository (don't initialize with README)
4. Click "Create repository"

Step 5: Connect and Push (if using website method)

1. Copy the repository URL from GitHub
2. In VS Code terminal, run:

```
git remote add origin [YOUR_REPO_URL]  
git branch -M main  
git push -u origin main
```

Option 2: Create Repository First, Then Clone

Step 1: Create GitHub Repository

1. Go to github.com and sign in
2. Click the "+" icon → "New repository"
3. Name your repository
4. Check "Add a README file" (optional)
5. Click "Create repository"

Step 2: Clone to Your Desired Location

1. Click the green "Code" button on your repository
2. Copy the HTTPS URL
3. In VS Code, open Command Palette (`Ctrl+Shift+P`)
4. Type "Git: Clone" and select it
5. Paste the URL and choose where to save it
6. Open the cloned folder in VS Code

Step 3: Add Your Files and Push

1. Copy your project files into the cloned folder
2. In VS Code Source Control panel (`Ctrl+Shift+G`):
 - Stage changes (click "+" next to files)
 - Write commit message
 - Click "Commit"
 - Click "Sync Changes" to push

Troubleshooting: Reset Remote Repository

Check Current Status

```
git status
git remote -v
```

Remove Existing Remote (if needed)

```
git remote remove origin
```

Re-establish Connection

```
git remote add origin [NEW_REPO_URL]
git branch -M main
git push -u origin main
```

Remove Git Entirely from Folder

Warning: This removes ALL git history and tracking

```
# Windows
rmdir /s .git

# Mac/Linux
rm -rf .git
```

After removing `.git`, you can start fresh with `git init`

Quick Tips

- **VS Code Git Panel:** Use `Ctrl+Shift+G` to access source control
- **Terminal Commands:** `Ctrl+``` opens integrated terminal
- **Authentication:** You may need to sign in to GitHub through VS Code
- **Branch Protection:** Main branch is typically protected in organizations